

GARRETT RUSSELL

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EDUCATION

Bachelor of Electrical Engineering, Purdue University, West Lafayette, IN

Aug 2024 - May 2028

GPA: 4.0/4.0

Coursework: ASIC Design, Signals and Systems, Microprocessor Systems, Electrical Engineering 1/2, Digital Design

Honors/Awards: NSCF Keynote Scholarship Finalist (top 2.5%), Dean's List (4x), ECE Scholarship

SKILLS

Technical Skills: PCBA/Schematic Design, Digital/Analog Design, Bucks, Flybacks, FPGAs, Soldering, Multimeter, EMI/EMC, Radiation, SAR ADC, Loop Analysis/Injection, Oscilloscope, Logic Analyzer, Monte Carlo
Software: Altium, KiCad, SPICE/LTspice, Cadence, Siemens TIA, EPlan, Git, CAD, Claude Code/Codex
Languages: SystemVerilog, C++, C, R, Python, MATLAB, LaTeX, JavaScript, HTML/CSS, SCL, Ladder Logic

EXPERIENCE

Electrical Engineering Intern

May 2026 - Aug 2026

Reflect Orbital

Los Angeles, CA

- Owned V3 Flight Computer trades, schematic, layout, routing, and bring-up while cutting size by 35% from prior gen (includes 3 Bucks, a Flyback, Load Switches, Ethernet, and 3 constant current drivers on an 8-layer EMC stackup)
- Designed, simulated, and tuned a 15W constant-current magnetorquer driver that cut EMI by removing high dv/dt edges
- Created and simulated a 60W 28V-28V Flyback Converter for isolated power in a custom radio
- Led next-gen compute trade study driven by radiation-tolerant design (mitigating TID, latchups, and SEUs)
- Authored guides for PCB standards, sizing input filters, FETs for linear operation, and standardized drawing templates

Analog Design Engineer

Jan 2026 - May 2026

Purdue SoCET

West Lafayette, IN

- Created a successive-approximation-register ADC for energy-critical applications, reducing power consumption by >60%
- Designed and optimized a Strong-Arm latch comparator using Cadence Virtuoso
- Developed the SAR algorithm for 10 MS/s 4-bit successive approximation in SystemVerilog and simulated performance

Automation and Controls Engineering Intern

Jan 2025 - Aug 2025

SpaceX

Brownsville, TX

- Designed entire hardware and electronics suite (4 custom panels/harnesses) for an auto-leveling system for vehicle transports
- Developed, simulated, and tested a 4-point leveling control algorithm to increase stability margin by >40%
- Owned software for a single-axis weldment rotator from ideation to production and deployed across 5+ cells factory-wide
- Created and troubleshoot production automation tools (electrical panels, PCBs, PLCs, Siemens Drives)

Undergraduate Data Science Researcher (Corporate Partners)

Aug 2024 - Dec 2024

The Data Mine, Purdue University

West Lafayette, IN

- Developed a customizable Monte Carlo simulation to accurately predict orbital collisions in Python with sgp4 and Matplotlib
- Collaborated with an interdisciplinary team under Viasat to determine a sustainable amount of space debris

PROJECTS

High Voltage Capacitor Charger

- Created a Mazzilli ZVS flyback driver for rapidly charging a 15,000uF 160VDC Capacitor bank

18650 Battery Assembler

- Trained robotic arm to assemble 18650 cells using teleoperated data + ACT policy (Honorable Best AMD Hack @StarkHacks)

Thrust Vector Control (TVC) Model Rocket

- Designed an STM32-based flight computer (UWB, RF comms, 3x pyro, 5V/8A buck) that is 70% smaller than prototypes
- Wrote full flight software stack (sensor drivers, Kalman filter, state machine, control loop)
- Designed, simulated, and flew a direct quaternion non-linear $P^2 + I$ controller that held the vehicle attitude within $<5^\circ$ of setpoint on the final of 4 launches across 2 design iterations
- Published build and how-to content (static-fire tests, control methods) reaching 1,000,000+ views across platforms